# Experiment No. 2

## Title

Apply CSS3 styling using Flexbox and Grid layout techniques on an existing HTML web page.

## Aim

To apply CSS3 styling techniques and implement Flexbox and Grid layouts on the HTML web page created in Experiment No. 1 for improving layout and visual presentation.

## Software Required

- Text Editor (Notepad / VS Code)
- Web Browser (Chrome / Firefox)

## Theory

CSS (Cascading Style Sheets) is a style sheet language used to control the presentation of HTML documents. While HTML is responsible for defining the structure and content of a web page, CSS controls how the content appears on the screen, including colors, fonts, spacing, alignment, and layout.

CSS3 introduces advanced layout modules such as **Flexbox** and **Grid**, which allow developers to design modern, responsive, and well-structured web pages. These layout techniques help in aligning elements efficiently and organizing content logically.

CSS (Cascading Style Sheets) is a stylesheet language used to control the presentation of web pages written in HTML. While HTML defines the structure and content of a web page, CSS defines how the content should appear on the screen, including layout, colors, fonts, spacing, and alignment.

CSS3 is the latest version of CSS and provides advanced features such as flexible layouts, animations, transitions, and responsive design. By separating content from presentation, CSS improves maintainability, consistency, and reusability of web pages.

## CSS Application Methods

CSS can be applied to HTML documents using:

1. **Inline CSS** – applied directly inside an HTML tag

2. **Internal CSS** – defined inside the `<style>` tag in the `<head>` section
3. **External CSS** – written in a separate `.css` file and linked to HTML

In this code fragment, **internal CSS** is used.

**Program Code**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="description" content="HTML Basics Demonstration">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML All Tags Demo</title>

  <!-- CSS Styling -->
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f0f4f8;
      margin: 0;
      line-height: 1.6;
    }

    header {
      background-color: #1e3a8a;
      color: white;
      padding: 20px;
      text-align: center;
    }

    nav {
      background-color: #2563eb;
      padding: 10px;
      text-align: center;
    }

    nav a {
      color: white;
      text-decoration: none;
      margin: 0 10px;
      font-weight: bold;
    }

    nav a:hover {
```

```
    text-decoration: underline;
}

section {
    background-color: white;
    margin: 20px;
    padding: 20px;
    border-radius: 8px;
}

h2 {
    color: #1e40af;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 10px;
}

table th {
    background-color: #bfdbfe;
}

table, th, td {
    border: 1px solid #ccc;
    padding: 10px;
    text-align: center;
}

img {
    border: 2px solid #cbd5e1;
    padding: 8px;
    border-radius: 10px;
    background-color: #f8fafc;
}

form input, form select {
    width: 100%;
    padding: 8px;
    margin-top: 5px;
}
```

```
        input[type="radio"],
        input[type="checkbox"] {
            width: auto;
        }

        input[type="submit"],
        input[type="reset"] {
            width: auto;
            padding: 10px 15px;
            margin-top: 10px;
            background-color: #2563eb;
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
        }

        input[type="reset"] {
            background-color: #6b7280;
        }

        footer {
            background-color: #1e3a8a;
            color: white;
            text-align: center;
            padding: 10px;
        }

        hr {
            border: none;
            height: 1px;
            background-color: #cbd5e1;
            margin: 20px;
        }
    </style>
</head>

<body>

<header>
    <h1>HTML Tags Demonstration</h1>
    <p><b>HTML</b> is used to create <i>web pages</i>.</p>
</header>
```

```
<nav>
   <a href="#lists">Lists</a>
   <a href="#table">Table</a>
   <a href="#media">Media</a>
   <a href="#form">Form</a>
</nav>

<section id="lists">
   <h2>Lists</h2>

   <h3>Unordered List</h3>
   <ul>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript</li>
   </ul>

   <h3>Ordered List</h3>
   <ol>
      <li>Create HTML file</li>
      <li>Add content</li>
      <li>Run in browser</li>
   </ol>

   <h3>Description List</h3>
   <dl>
      <dt>HTML</dt>
      <dd>Markup language for web pages</dd>
   </dl>
</section>

<section id="table">
   <h2>Table</h2>

   <table>
      <caption><b>Student Information</b></caption>
      <tr>
         <th>Roll No</th>
         <th>Name</th>
         <th>Branch</th>
      </tr>
      <tr>
```

```
            <td>1</td>
            <td>XYZ</td>
            <td>Computer</td>
        </tr>
    </table>
</section>

<section id="media">
    <h2>Image and Link</h2>

    <center>
        <img src="https://via.placeholder.com/150" alt="Sample Image">
        <br><br>
        <a href="https://www.w3schools.com" target="_blank">Learn HTML</a>
    </center>
</section>

<section id="form">
    <h2>Form</h2>

    <form>
        <fieldset>
            <legend><b>Registration Form</b></legend>

            Name:
            <input type="text" placeholder="Enter name" required>

            Email:
            <input type="email" required>

            Password:
            <input type="password">

            Gender:
            <input type="radio" name="gender"> Male
            <input type="radio" name="gender"> Female

            Course:
            <select>
                <option>Computer</option>
                <option>IT</option>
                <option>AIDS</option>
            </select>
```

```
        Skills:
        <input type="checkbox"> HTML
        <input type="checkbox"> CSS
        <input type="checkbox"> JS

        <br>
        <input type="submit" value="Register">
        <input type="reset">
      </fieldset>
    </form>
</section>

<footer>
    <p>&copy; 2026 Web Programming Lab</p>
</footer>

</body>
</html>
```

# Flexbox Layout

## Introduction

Flexbox (Flexible Box Layout) is a one-dimensional layout model used to arrange elements either in a row or a column. It is best suited for aligning items, distributing space, and creating flexible layouts.

Flexbox is commonly used for components such as navigation menus, toolbars, and form layouts.

## Important Flexbox Properties

- **display: flex** – enables Flexbox layout
- **flex-direction** – defines row or column arrangement
- **justify-content** – aligns items along the main axis
- **align-items** – aligns items along the cross axis
- **gap** – controls spacing between items

**Different Layouts of Flexbox:**

## 1. Horizontal (Row) Layout

```
display: flex;
flex-direction: row;
```

**Use case:** Navigation bars, menus
Items arranged left to right

### 2. Vertical (Column) Layout
display: flex;
flex-direction: column;
**Use case:** Forms, page sections
Items stacked top to bottom

### 3. Centered Layout
display: flex;
justify-content: center;
align-items: center;
**Use case:** Login boxes, cards
Content centered horizontally & vertically

### 4. Space Distribution Layout
justify-content: space-between;
**Use case:** Toolbars
Equal spacing between items

### 5. Wrapping Layout
flex-wrap: wrap;
**Use case:** Image galleries, cards
Items wrap to next line when space is less

**HTML File: (index.html)**

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="description" content="HTML Basics Demonstration">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>HTML All Tags Demo</title>

    <!-- External CSS -->
    <link rel="stylesheet" href="style.css">
</head>

<body bgcolor="#f0f4f8">

<header align="center">
```

```
    <h1>HTML Tags Demonstration</h1>
    <p><b>HTML</b> is used to create <i>web pages</i>.</p>
</header>

<hr>

<nav align="center">
    <a href="#lists">Lists</a> |
    <a href="#table">Table</a> |
    <a href="#media">Media</a> |
    <a href="#form">Form</a>
</nav>

<hr>

<section id="lists">
    <h2>Lists</h2>

    <h3>Unordered List</h3>
    <ul>
        <li>HTML</li>
        <li>CSS</li>
        <li>JavaScript</li>
    </ul>

    <h3>Ordered List</h3>
    <ol>
        <li>Create HTML file</li>
        <li>Add content</li>
        <li>Run in browser</li>
    </ol>

    <h3>Description List</h3>
    <dl>
        <dt>HTML</dt>
        <dd>Markup language for web pages</dd>
    </dl>
</section>

<hr>

<section id="table">
    <h2>Table</h2>
```

```
    <table border="1" cellpadding="10" cellspacing="0" align="center" width="60%">
        <caption><b>Student Information</b></caption>
        <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Branch</th>
        </tr>
        <tr>
            <td align="center">1</td>
            <td align="center">XYZ</td>
            <td align="center">Computer</td>
        </tr>
    </table>
</section>

<hr>

<section id="media">
    <h2>Image and Link</h2>

    <center>
        <img src="https://via.placeholder.com/150" alt="Sample Image">
        <br><br>
        <a href="https://www.w3schools.com" target="_blank">Learn HTML</a>
    </center>
</section>

<hr>

<section id="form">
    <h2>Form</h2>

    <form align="center">
        <fieldset>
            <legend><b>Registration Form</b></legend>

            Name:
            <input type="text" placeholder="Enter name" required><br><br>

            Email:
            <input type="email" required><br><br>
```

```
        Password:
        <input type="password"><br><br>

        Gender:
        <input type="radio" name="gender"> Male
        <input type="radio" name="gender"> Female<br><br>

        Course:
        <select>
          <option>Computer</option>
          <option>IT</option>
          <option>AIDS</option>
        </select><br><br>

        Skills:
        <input type="checkbox"> HTML
        <input type="checkbox"> CSS
        <input type="checkbox"> JS<br><br>

        <input type="submit" value="Register">
        <input type="reset">
      </fieldset>
    </form>
</section>

<hr>

<footer align="center">
    <p>&copy; 2026 Web Programming Lab</p>
</footer>

</body>
</html>
```

## Part A:Flexbox Code:(style.css):

**Program Code**

```
/* General Page Styling */
body {
    font-family: Arial, sans-serif;
    margin: 0;
    display: flex;
    flex-direction: column;
    gap: 20px;
```

```
}

/* Header */
header {
    background-color: #1e3a8a;
    color: white;
    padding: 20px;
}

/* FLEXBOX NAVIGATION */
nav {
    display: flex;
    justify-content: center;
    gap: 20px;
    background-color: #2563eb;
    padding: 10px;
}

nav a {
    color: white;
    text-decoration: none;
    font-weight: bold;
}

nav a:hover {
    text-decoration: underline;
}

/* Sections styled as cards */
section {
    background-color: white;
    margin: 0 20px;
    padding: 20px;
    border-radius: 8px;
}

/* Table Styling */
table {
    margin: auto;
    border-collapse: collapse;
```

```
}

th {
    background-color: #bfdbfe;
}

/* Image Styling */
img {
    border: 2px solid #cbd5e1;
    padding: 8px;
    border-radius: 10px;
}

/* Form Styling using Flexbox */
form {
    display: flex;
    flex-direction: column;
    align-items: center;
}

input, select {
    padding: 6px;
    margin-top: 5px;
}

input[type="submit"],
input[type="reset"] {
    padding: 8px 15px;
    margin-top: 10px;
    background-color: #2563eb;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}

input[type="reset"] {
    background-color: #6b7280;
}
```

```
/* Footer */
footer {
   background-color: #1e3a8a;
   color: white;
   padding: 10px;
}
```

# Grid Layout

## Introduction

CSS Grid is a two-dimensional layout system used to design web pages using rows and columns. It allows precise control over layout structure and is ideal for designing full-page layouts, dashboards, and multi-section pages.

Grid layout provides better control when both horizontal and vertical placement of elements is required.

**Important Grid Properties**

- **display: grid** – enables Grid layout
- **grid-template-columns** – defines column structure
- **grid-template-rows** – defines row structure
- **gap** – defines space between rows and columns

**Different Layouts of Grid:**

**1. Fixed Column Grid**
grid-template-columns: 1fr 1fr;
**Use case:** Simple 2-column layout

**2. 2×2 Box Layout**
grid-template-columns: repeat(2, 1fr);
**Use case:** Dashboard-style layouts
Lists | Table
Media | Form

**3. Unequal Column Layout**
grid-template-columns: 2fr 1fr;
**Use case:** Sidebar + content

**4. Auto-Fit Responsive Grid**

grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
**Use case:** Responsive cards
Adjusts automatically to screen size

**5. Full-Width Spanning Layout**
grid-column: 1 / -1;
**Use case:** Header, footer
Spans across all columns

**HTML File: (index.html)**

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="description" content="HTML Basics Demonstration">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>HTML All Tags Demo</title>

   <!-- External CSS -->
   <link rel="stylesheet" href="grid.css">
</head>

<body bgcolor="#f0f4f8">

<header align="center">
   <h1>HTML Tags Demonstration</h1>
   <p><b>HTML</b> is used to create <i>web pages</i>.</p>
</header>

<nav align="center">
   <a href="#lists">Lists</a> |
   <a href="#table">Table</a> |
   <a href="#media">Media</a> |
   <a href="#form">Form</a>
</nav>

<section id="lists">
   <h2>Lists</h2>
   <ul>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript</li>
   </ul>
</section>
```

```html
<section id="table">
   <h2>Table</h2>
   <table border="1" cellpadding="10" cellspacing="0" align="center">
      <tr>
         <th>Roll No</th>
         <th>Name</th>
         <th>Branch</th>
      </tr>
      <tr>
         <td>1</td>
         <td>XYZ</td>
         <td>Computer</td>
      </tr>
   </table>
</section>

<section id="media">
   <h2>Media</h2>
   <center>
      <img src="https://via.placeholder.com/150" alt="Sample Image">
      <br><br>
      <a href="https://www.w3schools.com" target="_blank">Learn HTML</a>
   </center>
</section>

<section id="form">
   <h2>Form</h2>
   <form align="center">
      <input type="text" placeholder="Name" required><br><br>
      <input type="email" placeholder="Email" required><br><br>
      <input type="submit" value="Register">
   </form>
</section>

<footer align="center">
   <p>&copy; 2026 Web Programming Lab</p>
</footer>

</body>
</html>
```

**PART B: GRID LAYOUT (CSS)**

**CSS Code – Grid (style.css)**

```
/* Page setup */
```

```
body {
    font-family: Arial, sans-serif;
    margin: 0;
    padding: 20px;

    display: grid;
    grid-template-columns: repeat(2, 1fr);
    grid-template-rows: auto auto auto;
    gap: 20px;
}

/* Header, Nav, Footer full width */
header,
nav,
footer {
    grid-column: 1 / -1;
    background-color: #1e3a8a;
    color: white;
    padding: 15px;
    text-align: center;
}

/* Navigation links */
nav a {
    color: white;
    text-decoration: none;
    margin: 0 10px;
    font-weight: bold;
}

/* 2x2 BOX SECTIONS */
section {
    background-color: white;
    padding: 20px;
    border-radius: 8px;
    border: 1px solid #d1d5db;
}

/* Headings */
h2 {
```

```
    text-align: center;
    color: #1e40af;
}

/* Table styling */
table {
    margin: auto;
    border-collapse: collapse;
}

/* Image styling */
img {
    border: 2px solid #cbd5e1;
    padding: 8px;
    border-radius: 8px;
}
```

**Student Tasks:**

**Task 1: Apply External CSS**

- Create an external CSS file.
- Link the CSS file to the existing HTML page created in Experiment No. 1.
- Use your creativity to apply styles of your choice.

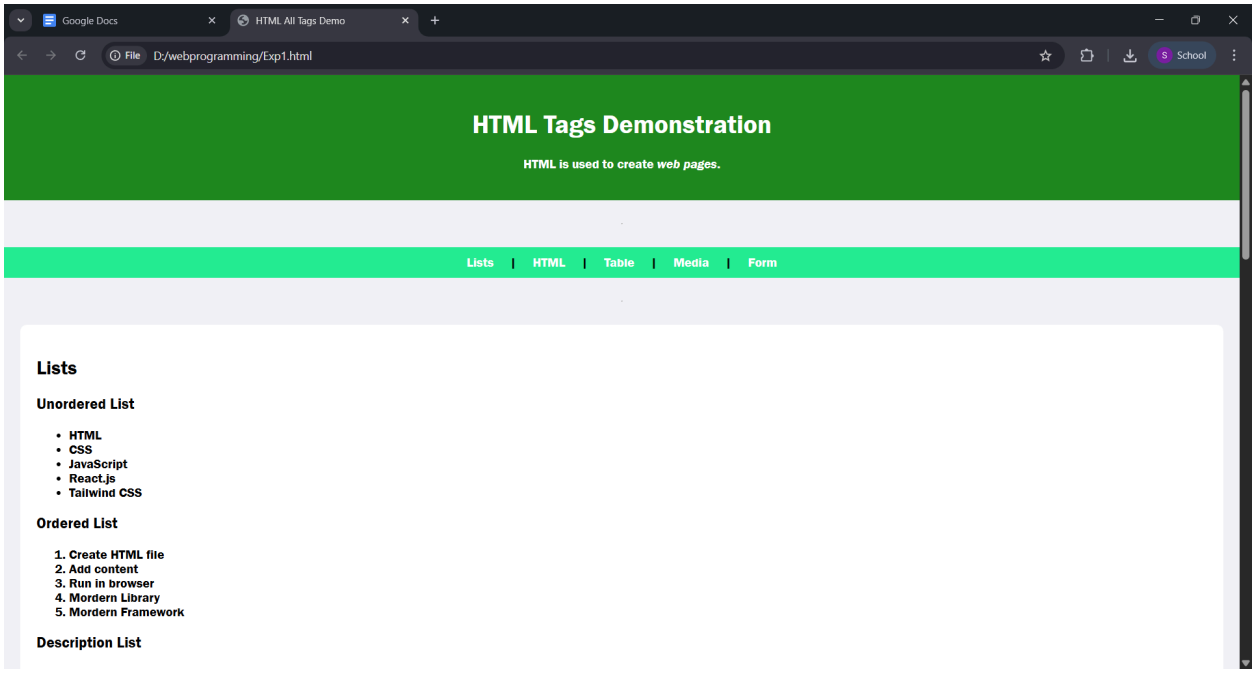
**Program:**

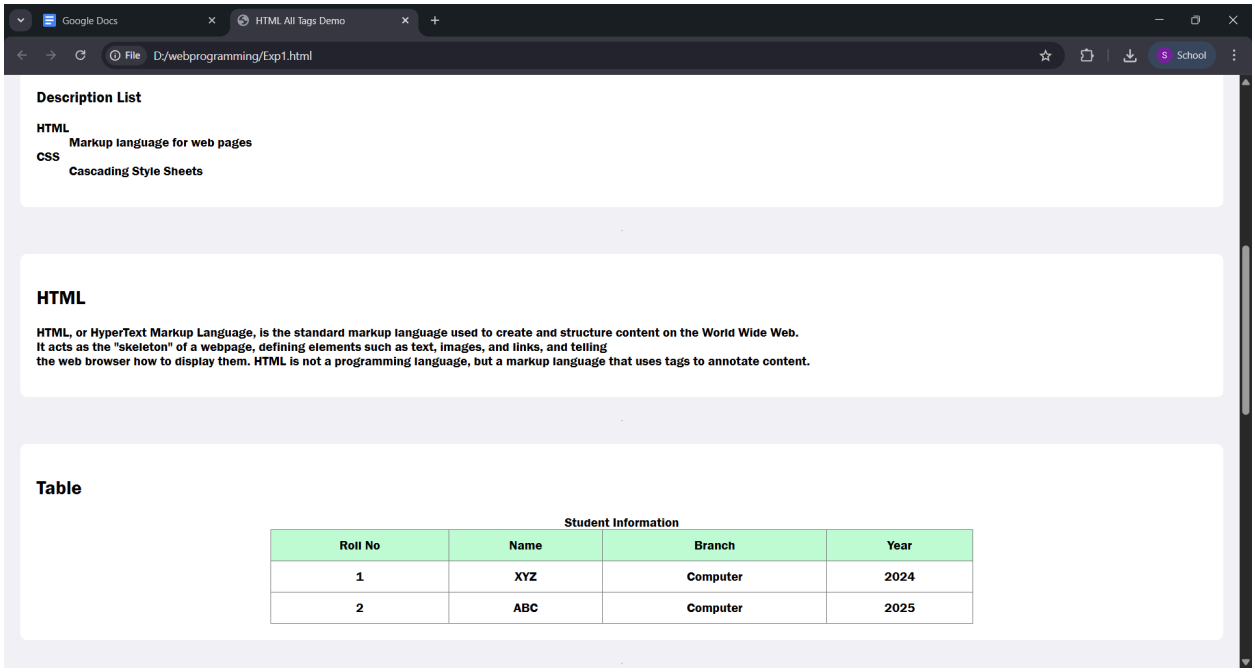
```
body{
    font-family: 'Franklin Gothic Medium', 'Arial Narrow', Arial, sans-serif, sans-serif;
    margin: 0;
    display: flex;
    flex-direction: column;
    gap: 20px;
}
header {
    background-color: #1e8a20;
    color: white;
    padding: 20px;
    align-items: center;
}
nav {
    display: flex;
    justify-content: center;
    gap: 20px;
    background-color: #25eb92;
```

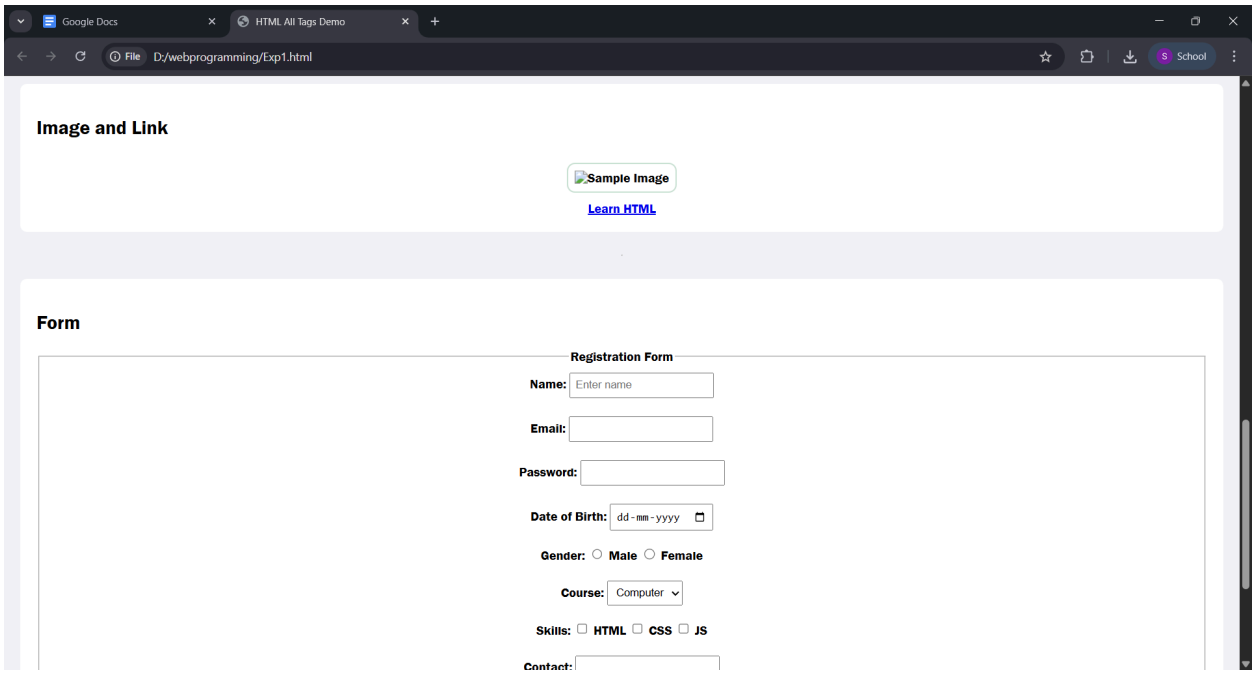
```
    padding: 10px;
}
nav a {
    color: white;
    text-decoration: none;
    font-weight: bold;
}

nav a:hover {
    text-decoration: underline;
}
section {
    background-color: white;
    margin: 0 20px;
    padding: 20px;
    border-radius: 8px;
}
table {
    margin: auto;
    border-collapse: collapse;
}

th {
    background-color: #bffed3;
}
img {
    border: 2px solid #cbe1d5;
    padding: 8px;
    border-radius: 10px;
}
form {
    display: flex;
    flex-direction: column;
    align-items: left;
}

input, select {
    padding: 6px;
    margin-top: 5px;
}

input[type="submit"],
input[type="reset"] {
```

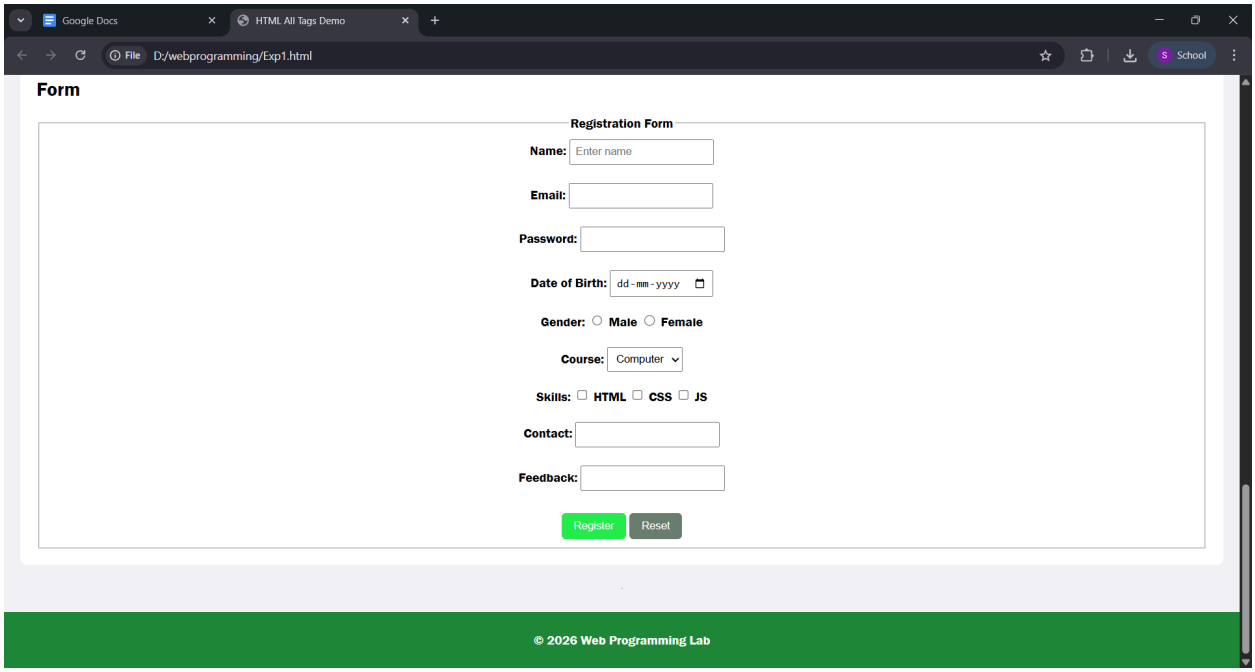
```
    padding: 8px 15px;
    margin-top: 10px;
    background-color: #25eb4d;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}

input[type="reset"] {
    background-color: #6b8070;
}

footer {
    background-color: #1e8a39;
    color: white;
    padding: 10px;
}
```

**Output:**




Google Docs
HTML All Tags Demo
File D:/webprogramming/Exp1.html
School
Description List
HTML
Markup language for web pages
CSS
Cascading Style Sheets
HTML
HTML, or HyperText Markup Language, is the standard markup language used to create and structure content on the World Wide Web.
It acts as the "skeleton" of a webpage, defining elements such as text, images, and links, and telling
the web browser how to display them. HTML is not a programming language, but a markup language that uses tags to annotate content.
Table
Student Information
Roll No | Name | Branch | Year
1 | XYZ | Computer | 2024
2 | ABC | Computer | 2025


Google Docs
HTML All Tags Demo
File D:/webprogramming/Exp1.html
School
Image and Link
Sample Image
Learn HTML
Form
Registration Form
Name: Enter name
Email:
Password:
Date of Birth: dd-mm-yyyy
Gender: Male Female
Course: Computer
Skills: HTML CSS JS
Contact:




**Task 2: Flexbox Layout Implementation**

- Apply **Flexbox** to the navigation bar.
- Explore different layouts of Flexbox

**Program:**

```
body {
   font-family: Arial, sans-serif;
   margin: 0;
   display: flex;
   flex-direction: column;
   gap: 20px;
}

/* Header */
header {
   background-color: #1e3a8a;
   color: white;
   padding: 20px;
}

/* FLEXBOX NAVIGATION */
nav {
   display: flex;
   justify-content: center;
   gap: 20px;
```

```
    background-color: #2563eb;
    padding: 10px;
}

nav a {
    color: white;
    text-decoration: none;
    font-weight: bold;
}

nav a:hover {
    text-decoration: underline;
}

/* Sections styled as cards */
section {
    background-color: white;
    margin: 0 20px;
    padding: 20px;
    border-radius: 8px;
}

/* Table Styling */
table {
    margin: auto;
    border-collapse: collapse;
}

th {
    background-color: #bfdbfe;
}

/* Image Styling */
img {
    border: 2px solid #cbd5e1;
    padding: 8px;
    border-radius: 10px;
}

/* Form Styling using Flexbox */
form {
    display: flex;
    flex-direction: column;
```

```
    align-items: center;
}

input, select {
    padding: 6px;
    margin-top: 5px;
}

input[type="submit"],
input[type="reset"] {
    padding: 8px 15px;
    margin-top: 10px;
    background-color: #2563eb;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}

input[type="reset"] {
    background-color: #6b7280;
}

/* Footer */
footer {
    background-color: #1e3a8a;
    color: white;
    padding: 10px;
}
```

**Output:**

HTML All Tags Demo

File D:/webprogramming/Exp2.html

# HTML Tags Demonstration

**HTML** is used to create *web pages*.

Lists Table Media Form

## Lists

### Unordered List

- HTML
- CSS
- JavaScript

### Ordered List

1. Create HTML file
2. Add content
3. Run in browser

### Description List

HTML

HTML All Tags Demo

File D:/webprogramming/Exp2.html

### Description List

HTML
Markup language for web pages

## Table

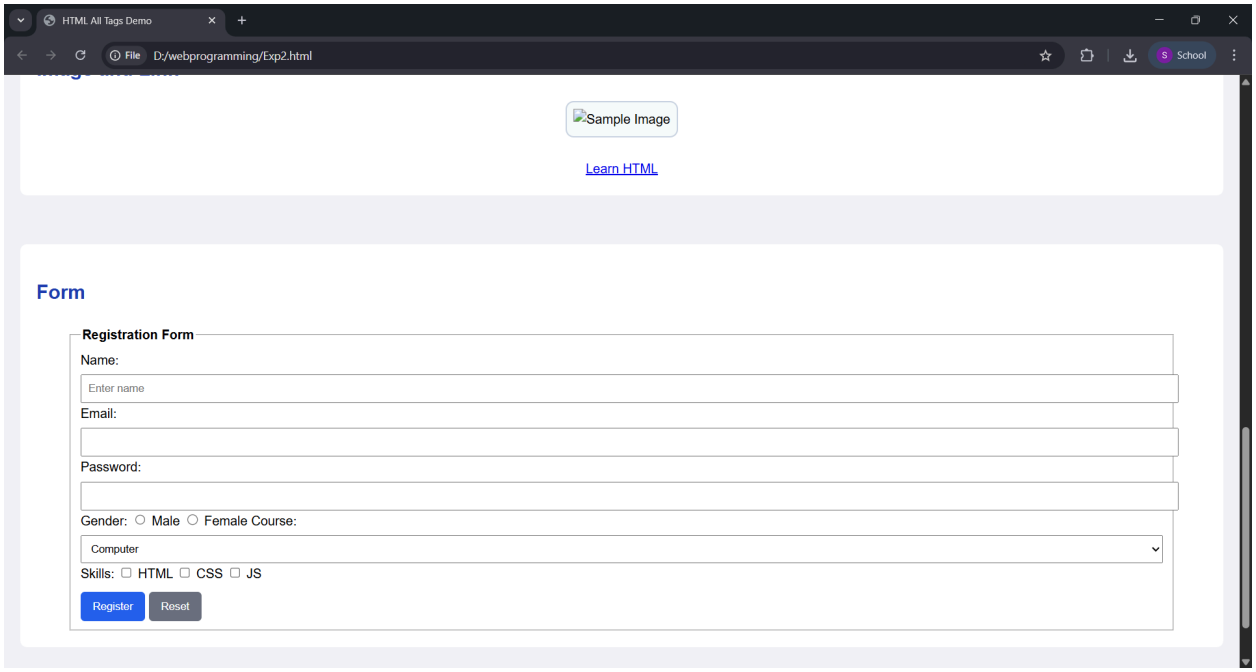
**Student Information**

| Roll No | Name | Branch |
|---|---|---|
| 1 | XYZ | Computer |

## Image and Link

Sample Image

Learn HTML




**Task 4: Grid Layout Implementation**

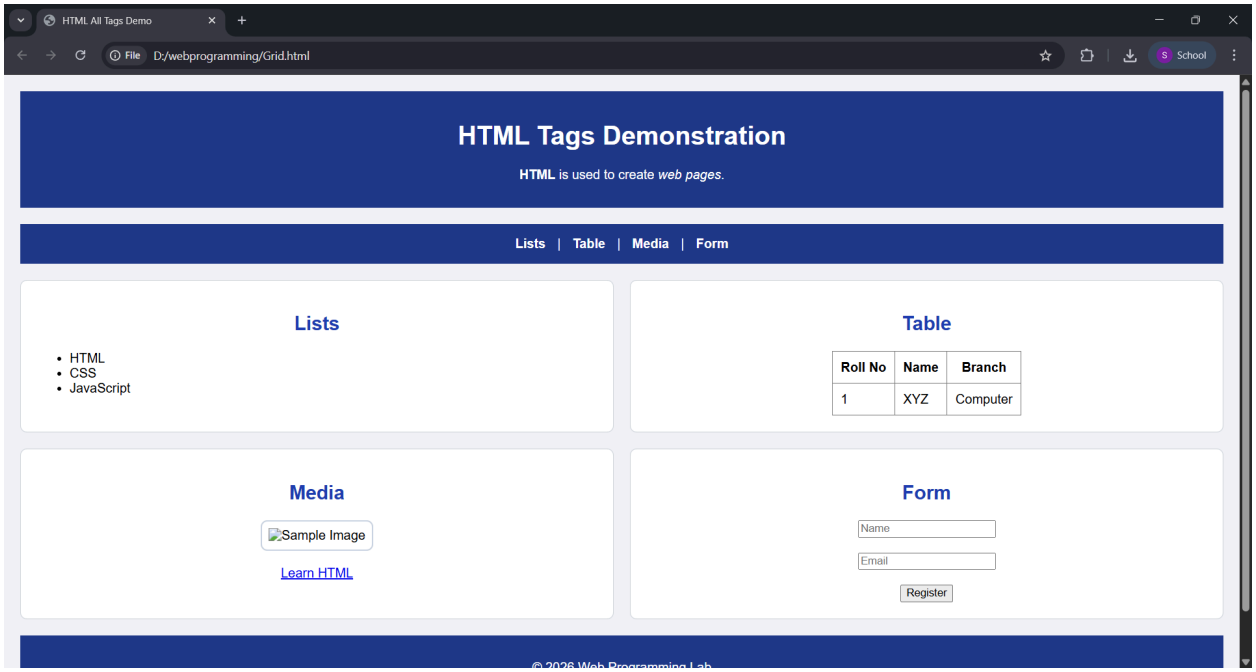
- Explore different layouts of grid.

**Program:**

```
body {
   font-family: Arial, sans-serif;
   margin: 0;
   padding: 20px;

   display: grid;
   grid-template-columns: repeat(2, 1fr);
   grid-template-rows: auto auto auto;
   gap: 20px;
}

/* Header, Nav, Footer full width */
header,
nav,
footer {
   grid-column: 1 / -1;
   background-color: #1e3a8a;
   color: white;
   padding: 15px;
   text-align: center;
}

/* Navigation links */
```

```
nav a {
    color: white;
    text-decoration: none;
    margin: 0 10px;
    font-weight: bold;
}

/* 2x2 BOX SECTIONS */
section {
    background-color: white;
    padding: 20px;
    border-radius: 8px;
    border: 1px solid #d1d5db;
}

/* Headings */
h2 {
    text-align: center;
    color: #1e40af;
}

/* Table styling */
table {
    margin: auto;
    border-collapse: collapse;
}

/* Image styling */
img {
    border: 2px solid #cbd5e1;
    padding: 8px;
    border-radius: 8px;
}
```

**Output:**




**Task 5: Comparison and Observation**

- Compare the layout created using Flexbox and Grid.
- Identify which layout technique is more suitable for:
    - Navigation alignment
    - Page structure with rows and columns
- Note down the observations.

Flexbox and Grid are both powerful CSS layout techniques, but they serve different purposes. Flexbox is a one-dimensional layout system that works best when elements need to be aligned in a single row or column, making it more suitable for navigation alignment where items can be easily spaced, centered, or distributed. Grid, on the other hand, is a two-dimensional layout system that allows designers to control both rows and columns at the same time, making it more suitable for creating overall page structures and complex layouts. From observation, Flexbox is simpler and more flexible for small UI components like navbars, while Grid provides better control and structure for full-page layouts. In practice, combining both Flexbox and Grid often results in the most efficient and responsive designs.

**Result**

All